Two-Stage Robust Planning for Urban Distribution System with Heatwave Seasonal Imbalance and Progressive Intensification

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***Abstract*—Irreversible global warming is intensifying urban heatwave events, leading to surges in cooling demand and reductions in generation efficiency, which pose severe source–load imbalance challenges for distribution systems. Hence, this paper proposes a two-stage robust planning model that incorporates the multi-timescale evolution of heatwaves. First, a hybrid meteorological modeling framework based on Seasonal Autoregressive Moving Average and Long Short-Term Memory Networks is developed to capture both short-term seasonal fluctuations and long-term evolving trends. Furthermore, a compound uncertainty set is constructed by integrating multiple meteorological factors. Second, the dynamic source–load response model is established for renewable generators and building cooling loads, reflecting the impacts of weather conditions on generation efficiency and thermal processes. Then, a two-stage robust planning model is formulated, where the first stage determines investment decisions for distributed resources, while the second stage evaluates worst-case operational costs under all admissible climate scenarios. To address high-dimensional uncertainty, a Benders decomposition approach is employed to improve computational tractability. Finally, Numerical studies on an urban distribution system demonstrate the effectiveness of the proposed model in balancing investment cost, operational performance, and robustness under compound heatwave stress.**

***Index Terms*—Urban heatwaves, distribution system, seasonal imbalance, progressive intensification, robust planning.**

I. Introduction

The accelerating global warming has led to frequent high temperature shocks around the world [1]. Particularly in urban areas, high population density, building structures, and energy activities cause temperatures in city centers to be significantly higher than in surrounding suburbs [2]. In recent years, major cities worldwide are experiencing persistent high-temperature episodes, called "urban heatwaves", whose intensity, frequency, and spatial coverage have been escalating year by year [3]. Prolonged heat exposure has triggered surging cooling demand, hydropower shortages, thermal generation derating, and equipment failures, ultimately resulting in widespread blackouts—such as the 2012 India blackout, the 2022 California rolling outages, and the 2024 Balkan Peninsula power crisis [4]. These events highlight the urgent need to incorporate urban heatwaves into long-term power system planning to enhance reliability.

Numerous studies have investigated the impact of meteorological events on power system planning, operation, and control, which can be broadly classified into three categories. *i)* The first aims to enhance system resilience against external shocks and improve operational flexibility through planning strategies. Representative methods include the deployment of storage and reserve units [5], network topology optimization [6], and the integration of flexible interconnection devices [7]. *ii)* The second employs advanced control and coordination strategies to enhance operational flexibility and recovery capabilities during extreme conditions, achieved by dispatching interruptible, deferrable, or curtailable loads [8] [9]. *iii)* The third concentrates on modeling the uncertainty of meteorological events, proposing mathematical methods including multi-scenario simulation [10], robust optimization, chance-constrained programming [11], and distributionally robust optimization [12]. However, most existing research primarily focuses on low-probability, high-impact events such as typhoons, floods, and ice storms. These events are typically modeled as discrete probability scenarios or tail risk problems, emphasizing emergency response and rapid recovery capabilities. In contrast, urban heatwaves represent a periodically recurring meteorological process exhibiting a trend of continuous intensification, characterized by prolonged duration and extensive coverage. Therefore, current research on heatwaves remains limited, lacking high-resolution climate modeling frameworks capable of capturing both short-term seasonal fluctuations and long-term intensification trends.

Heatwave events cause systemic disturbances across all segments of the distribution system [13]. Specifically, in the source side, the photovoltaic and wind generation are highly sensitive to meteorological conditions [14]. Prolonged heat conditions cause thermal derating of PV modules and increased variability in wind power output [15]. Network-side equipment faces dual pressures of high load and high temperature, leading to cable capacity exceeding limits [16].

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Meanwhile, building cooling demand is rapidly increasing due to rising ambient temperatures [17]. Therefore, these dynamics reveal a strong meteorology–power coupling during heatwave conditions. Prior studies have proposed dispatch strategies to mitigate peak load stress [18] and established dynamic numerical models for temperature-dependent facilities [19]. However, most existing models depend on single-point meteorological measurements or simplified forecast profiles. These approaches are unable to represent the multi-timescale uncertainties inherent in heatwave dynamics [20]. As a result, the dynamic impacts of heatwave evolution on source-load imbalance remain inadequately explored.

Therefore, this paper proposes a two-stage robust planning framework to enhance the heatwave resilience of distribution systems. The main contributions are as follows:

1) A multi-timescale meteorological modeling framework is developed by integrating Seasonal Autoregressive Moving Average models with Long Short-Term Memory Networks, enabling the characterization of both short-term seasonal imbalances and long-term progressive intensification of heatwaves.

2) Climate-responsive models are established to characterize the dynamic behavior of source and load components under heatwaves, including temperature-dependent efficiency degradation models for wind and photovoltaic generation, and thermal balance model for building cooling demand.

3) A two-stage robust optimization model is formulated to jointly optimize investment and operation decisions under heatwave-driven uncertainties. The Benders decomposition algorithm is applied to enhance computational efficiency.

The remainder of this paper is organized as follows. Section II introduces the meteorological modeling framework. Section III formulates the climate-responsive models for distribution system. Section IV presents the two-stage robust planning formulation. Case studies are given in Section V, followed by conclusions in Section VI.

II. Multi-timescale Evolution Model for Urban Heatwaves

In this section, the uncertain evolution of urban heatwaves is decomposed into short-term seasonal variations and long-term progressive intensification. A hybrid meteorological modeling framework based on Seasonal Autoregressive Moving Average (SARMA) and Long Short-Term Memory Networks (LSTM) is developed.

A. Seasonal Imbalance Model Based on SARMA

In the short-term evolution, urban heatwaves exhibit strong seasonal periodicity and intra-seasonal fluctuations. The SARMA model is developed to combine the temporal structure and seasonal cyclical variations. First, let $\mathbf{Y}_t \in \mathbb{R}^k$ denote the vector of k meteorological variables at time step t, as shown below:

$$\mathbf{Y}_t = \begin{bmatrix} T_t & V_t & R_t & \cdots \end{bmatrix}^T \tag{1}$$

where T_t, V_t, and R_t are the ambient temperature, wind speed, and solar irradiance, respectively. We adopt the standard seasonal difference-integrated SARMA model with seasonal period s ($s = 24$ for hourly data with daily seasonality). The multivariate SARMA model is formulated as:

$$\boldsymbol{\Phi}(B^s)\boldsymbol{\phi}(B)\nabla^d\nabla_s^D\mathbf{Y}_t = \boldsymbol{\Theta}(B^s)\boldsymbol{\theta}(B)\boldsymbol{\varepsilon}_t \tag{2}$$

where B is the lag operator; $\nabla = 1 - B$ is the regular difference operator, and ∇^d removes polynomial trend of degree d; $\nabla_s = 1 - B^s$ is the seasonal difference operator, and ∇_s^D removes seasonal trends of degree D; $\phi(B) = \mathbf{I} - \sum_{i=1}^{p}\phi_i B^i$ is the non-seasonal autoregressive operator matrix; $\boldsymbol{\theta}(B) = \mathbf{I} + \sum_{j=1}^{q}\boldsymbol{\theta}_j B^j$ is the non-seasonal moving average operator matrix; $\boldsymbol{\Phi}(B^s) = \mathbf{I} - \sum_{m=1}^{P}\boldsymbol{\Phi}_m B^{sm}$ is the seasonal AR operator matrix; $\boldsymbol{\Theta}(B^s) = \mathbf{I} + \sum_{n=1}^{Q}\boldsymbol{\Theta}_n B^{sn}$ is the seasonal MA operator matrix; $\boldsymbol{\varepsilon}_t \in \mathbb{R}^k$ is a zero-mean white noise vector with covariance matrix $\boldsymbol{\Sigma}_\varepsilon$, i.e., $\mathbb{E}[\varepsilon_t\varepsilon_t^T] = \boldsymbol{\Sigma}_\varepsilon$ and $\mathbb{E}[\boldsymbol{\varepsilon}_t\boldsymbol{\varepsilon}_{t-h}^T] = \mathbf{0}$ for $h \neq 0$. The model parameters $\{\phi_i, \Phi_m, \theta_j, \Theta_n\}$ and the noise covariance matrix $\boldsymbol{\Sigma}_\varepsilon$ are estimated using Maximum Likelihood Estimation ethods, based on historical multi-year meteorological time series.

B. Long-Term Progressive Intensification Model

The LSTM model is designed to capture the long-term evolutionary trends of heatwaves, reflecting the gradual increase in temperature due to climate change. The LSTM network consists of a series of memory cells designed to preserve long-term dependencies, governed by the following steps:

1) Forget Gate: This gate determines which information from the previous time step should be discarded.

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \tag{3}$$

where f_t is the forget gate; σ is the sigmoid activation function; W_f is the wight matrices; h_{t-1} is the cell state vector at time step $t-1$; x_t is the input of the neuron at time step t; b_f is the bias term.

2) Input Gate: This gate decides which new information should be added to the memory cell.

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \tag{4}$$

where i_t is the input gate; W_i and b_i are the wight matrices and bias term, respectively.

3) Cell State Update: The new memory is updated based on the previous memory and the current input.

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \tag{5}$$

$$C_t = f_t \cdot C_{t-1} + i_t \cdot \tilde{C}_t \tag{6}$$

where C_t is the cell state; $\tanh$ is the hyperbolic tangent function; W_C and b_C are the wight matrices and bias term.

4) Output Gate: This gate determines the output based on the current memory and the current input.

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \tag{7}$$

$$h_t = o_t \cdot \tanh(C_t) \tag{8}$$

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where o_t is the Output gate; W_o and b_o are the wight matrices and bias term, respectively.

C. Model Fusion and Compound Uncertainty Set Generation

The prediction of meteorological parameters is obtained by combining the outputs of SARMA and LSTM models:

$$\hat{\mathbf{Y}}_t = \hat{\mathbf{Y}}_t^{\text{SARMA}} + \hat{\mathbf{Y}}_t^{\text{LSTM}} \tag{9}$$

where $\hat{\mathbf{Y}}_t^{\text{SARMA}}$ and $\hat{\mathbf{Y}}_t^{\text{LSTM}}$ are the predicted output from the SARMA and LSTM model, respectively.

To incorporate the multi-timescale uncertainty of heatwaves, a robust compound uncertainty set is constructed based on the joint output of the SARMA–LSTM hybrid model. Let $\hat{\mathbf{Y}}_t^{(n)} \in \mathbb{R}^k$ denote the $n-$th synthetic realization of meteorological parameters at time t. Based on (9), the robust uncertainty set $\mathcal{U}_{\text{metco}} \in \mathbb{R}^k$ is defined as the convex hull of all generated meteorological realizations:

$$\mathcal{U}_{\text{metco}} = \left\{ \mathbf{Y}_t \in \mathbb{R}^k | \mathbf{Y}_t = \sum_{n=1}^{N} \lambda_n \hat{\mathbf{Y}}_t^{(n)} \right\} \tag{10}$$

where $\lambda_n \geq 0$ and $\sum_{n=1}^{N} \lambda_n = 1$. This polyhedral set forms a compact and convex representation of all plausible meteorological realizations observed from the scenario generation process. It preserves both the temporal structure and statistical diversity across the full space of high-temperature weather evolution under multi-timescale uncertainty.

III. Climate-Driven Dynamic Operation Model of Distribution System Components

To capture the heatwave-induced operational response of distribution systems, this section establishes physics-informed models for critical distribution-level components: building cooling demand, photovoltaic and wind power generation. All models link component behavior to meteorological inputs and support spatial mapping to buses via geospatial association.

A. Building Cooling Load

Urban buildings are thermally excited by high ambient temperatures and solar radiation during heatwaves, causing cooling loads to vary dynamically. The internal temperature $T_{in}^{(b)}(t)$ of building b evolves under the dynamic heat balance:

$$\begin{aligned} C_{th}^{(b)} \cdot \frac{dT_{in}^{(b)}(t)}{dt} &= \dot{Q}_{cond}^{(b)}(t) + \dot{Q}_{conv}^{(b)}(t) + \dot{Q}_{rad}^{(b)}(t) \\ &+ \dot{Q}_{int}^{(b)}(t) - \dot{Q}_{cood}^{(b)}(t) \end{aligned} \tag{11}$$

where $C_{th}^{(b)}$ is the effective thermal capacitance of building b; $\dot{Q}_{cond}^{(b)}(t)$ is the heat conduction through walls/roof; $\dot{Q}_{rad}^{(b)}(t)$ is the convective heat from ambient air; $\dot{Q}_{rad}^{(b)}(t)$ is the solar radiation through windows; $\dot{Q}_{int}^{(b)}(t)$ is the internal heat gains from people and appliances; $\dot{Q}_{cood}^{(b)}(t)$ is the cooling extraction via air conditioners. Each Heat gain term is detailed as follows:

$$\dot{Q}_{cond}^{(b)}(t) = \sum_{s \in \mathcal{S}_b} U_s^{(b)} \cdot A_s^{(b)} \cdot (T_{amb}(t) - T_{in}^{(b)}(t)) \tag{12}$$

$$\dot{Q}_{rad}^{(b)}(t) = \sum_{w \in \mathcal{W}_b} \alpha_w^{(b)} \cdot A_w^{(b)} \cdot R_t \tag{13}$$

$$\dot{Q}_{int}^{(b)} = N_{occ}^{(b)} \cdot q_{occ} + q_{plug}^{(b)}(t) \tag{14}$$

$$\dot{Q}_{cood}^{(b)}(t) = P_{ac}^{(b)}(t) \cdot \text{COP}^{(b)} \tag{15}$$

where $\mathcal{S}_b$ is the set of envelope surfaces (walls, roof, etc.); $U_s^{(b)}$ and $A_s^{(b)}$ are the thermal transmittance and area of surface s, respectively; $T_{amb}(t)$ is the ambient temperature at time t; $\mathcal{W}_b$ is the set of window surfaces in building b; $\alpha_w^{(b)}$ and $A_w^{(b)}$ are the solar heat gain coefficient and area of window w, respectively; R_t is solar irradiance at time t; $N_{occ}^{(b)}$ is number of occupants; q_{occ} and $q_{plug}^{(b)}(t)$ are the heat gain per occupant and plug/appliance loads, respectively; $P_{ac}^{(b)}(t)$ and $\text{COP}^{(b)}$ are the power and coefficiency of performane of air conditioner.

B. Dynamic PV Output Model

During heatwave events, photovoltaic (PV) output is affected not only by solar irradiance but also by panel temperature, which is governed by transient thermal exchange with the surrounding environment. The temperature of the PV module $T_{mod}(t)$ evolves according to a lumped thermal balance:

$$C_{pv}^{(r)} \cdot \frac{dT_{mod}^{(r)}(t)}{dt} = \dot{Q}_{abs}^{(r)}(t) - \dot{Q}_{conv}^{(r)}(t) - \dot{Q}_{rad}^{(r)}(t) - P_{pv}^{(r)}(t) \tag{16}$$

where $C_{pv}^{(r)}$ is the effective thermal capacitance of PV module r; $\dot{Q}_{abs}^{(r)}(t)$ is the absorbed solar radiation; $\dot{Q}_{conv}^{(r)}(t)$ is the convective heat loss to ambient air; $\dot{Q}_{rad}^{(r)}(t)$ is the radiative heat loss to the sky; $P_{pv}^{(r)}(t)$ is the extracted electric power. Each Heat gain term is detailed as follows:

$$\dot{Q}_{abs}^{(r)}(t) = \alpha_{glass}^{(r)} \cdot A_{pv}^{(r)} \cdot R_t \tag{17}$$

$$\dot{Q}_{conv}^{(r)}(t) = h_{amb}^{(r)} \cdot A_{pv}^{(r)} \cdot (T_{mod}^{(r)}(t) - T_{amb}(t)) \tag{18}$$

$$\dot{Q}_{rad}^{(r)}(t) = \epsilon_{pv} \cdot \sigma \cdot A_{pv}^{(r)} \cdot (T_{mod}^{4(r)} - T_{sky}^{4}(t)) \tag{19}$$

$$P_{pv}^{(r)}(t) = \eta_{ref} \cdot (1 - \mu_T \cdot (T_{mod}^{(r)}(t) - T_{ref})) \cdot A_{pv}^{(r)} \cdot R_t \tag{20}$$

where α_{glass} is effective absorption coefficient; A_{pv} is active PV area; h_{amb} is the convective heat transfer coefficient; ϵ_{pv} is the emissivity; σ is the Stefan-Boltzmann constant; T_{sky} is the effective sky temperature, typically approximated as $T_{sky}(t) = T_{amb}(t) - \Delta T_{sky}$ with $\Delta T_{sky} \in [10, 20]°\text{C}$; η_{ref} is the reference efficiency at $T_{ref} = 25°\text{C}$.

C. Wind Power Output Model

Wind turbine output is governed primarily by wind speed at the hub height. The standard power curve of a wind turbine reflects three distinct operating regimes: below cut-in, partial load, and rated plateau, beyond which turbines shut down at cut-out speed for safety. The output power $P_w^{(g)}(t)$ of wind turbine g at time t is modeled as:

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$$P_w^{(g)}(t) = \begin{cases} 0, & V_t^{(g)} < V_{\text{ci}}, V_t^{(g)} \geq V_{\text{co}} \\ P_{\text{rated}}^{(g)} \cdot \left(\frac{\left(V_t^{(g)}\right)^3 - V_{\text{ci}}^3}{V_{\text{r}}^3 - V_{\text{ci}}^3} \right), & V_{\text{ci}} \leq V_t^{(g)} < V_{\text{r}} \\ P_{\text{rated}}^{(g)}, & V_{\text{r}} \leq V_t^{(g)} < V_{\text{co}} \end{cases} \tag{21}$$

where $V_t^{(g)}$ is the wind speed at hub height for turbine g; $P_{\text{rated}}^{(g)}$ is the rated power; V_{ci}, V_r, and V_{co} are the cut-in speed, rated speed, and cut-out speed, respectively.

D. Uncertainty Set Reformulation

Each building b, PV r, and wind power g are geolocated to a distribution node $i \in \mathcal{N}$.

$$P_{ac}^{(i)}(t) = \sum_{b \in \mathcal{B}_i} P_{ac}^{(b)}(t) \tag{22}$$

$$P_{pv}^{(i)}(t) = \sum_{r \in \mathcal{R}_i} P_{pv}^{(r)}(t) \tag{23}$$

$$P_w^{(i)}(t) = \sum_{g \in \mathcal{G}_i} P_w^{(g)}(t) \tag{24}$$

where $\mathcal{B}_i$, $\mathcal{R}_i$, and $\mathcal{G}_i$ denote the set of buildings, PV modules, and Wind turbines connected to node i. The net power injection at each distribution node $i \in \mathcal{N}$ is computed by aggregating all climate-responsive components:

$$P_{net}^{(i)}(t) = P_{pv}^{(i)}(t) + P_w^{(i)}(t) - P_{ac}^{(i)}(t) \tag{25}$$

Based on the SARMA-LSTM in Section II, we obtain a set of meteorological sample sequences representing the seasonal evolution and variability of heatwaves:

$$\left\{ \hat{\boldsymbol{\xi}}^{(n)} \right\}_{n=1}^{N} = \left\{ \left(\hat{T}_{amb}^{(n)}(t), \hat{V}^{(n)}(t), \hat{R}^{(n)}(t) \right)_{t \in \mathcal{T}} \right\} \tag{26}$$

Concatenating all node i, time t, and scenarios n, the high-dimensional net injection vector for the scenario is obtained:

$$\hat{\mathbf{P}}_{\text{net}}^{(n)} = \left[\hat{P}_{\text{net}}^{(1,n)}(t_1), \ldots, \hat{P}_{\text{net}}^{(|\mathcal{N}|,n)}(t_{|\mathcal{T}|}) \right]^T \tag{27}$$

Furthermore, based on the meteorologically driven scenario, construct the following convex hull-shaped uncertainty set:

$$\mathcal{U} := \text{Conv}\left(\left\{ \hat{\mathbf{P}}_{\text{net}}^{(n)} \right\}_{n=1}^{N} \right) \tag{28}$$

where $\mathcal{U}$ is the polyhedral uncertainty set; $\hat{\mathbf{P}}_{\text{net}}^{(n)}$ accounts for load variations, generation efficiency degradation, and renewable output volatility under heatwave conditions.

IV. Two-Stage Robust Planning Framework Considering Multi-Timescale Uncertainty

A. Two-Stage Robust Optimization Framework

To address climate-driven uncertainty in urban distribution systems, a two-stage robust planning model is formulated in this section. The first stage involves investment decisions including the siting and sizing of distributed resources. The second stage represents operational decisions under the worst-case realization of uncertain net power injections.

$$\min_{\mathbf{x}} \left\{ C_{\text{inv}}(\mathbf{x}) + \max_{\mathbf{P}_{\text{net}} \subset \mathcal{U}} \min_{\mathbf{y}} C_{\text{op}}(\mathbf{y} : \mathbf{x}, \mathbf{P}_{\text{net}}) \right\} \tag{29}$$

$$C_{\text{inv}}(\mathbf{x}) = \sum_{j \in \mathcal{J}} \left(c_j^{\text{inv}} s_j + c_j^{\text{fix}} z_j \right) \tag{30}$$

$$C_{\text{op}}(\mathbf{y}) = \sum_{t \in \mathcal{T}} \left(\sum_{g \in \mathcal{G}} c_g^{\text{gen}} P_g(t) + \sum_{i \in \mathcal{N}} c_i^{\text{shed}} LS_i(t) \right) \tag{31}$$

where C_{inv} and $C_{\text{op}}(\mathbf{y})$ are the investment cost and operational cost, respectively; $\mathbf{x}$ and $\mathbf{y}$ are the first-stage investment decisions (e.g., location and sizing of generation and storage) and second-stage operation decisions (e.g., output of controllable units), respectively; c_j^{inv} and c_j^{fix} are the per-unit and fixed installation costs, respectively; s_j and z_j are the installed capacity and binary variable indicating investment selection at site j; c_g^{gen} and c_i^{shed} are the generation costs and penalty costs for load shedding, respectively; $P_g(t)$ and $LS_i(t)$ are the generation and load shedding at node i, respectively.

B. Planning and Operational Constraints

1) Siting and Sizing Limits: These constraints define the installation limits of distributed energy resources and enforce binary investment decisions:

$$0 \leq s_j \leq \bar{s}_j z_j, \quad z_j \in \{0, 1\}, \quad \forall j \in \mathcal{J} \tag{32}$$

where $\bar{s}_j$ is the maximum allowed installation capacity.

2) Power Flow Constraints: To represent nonlinear radial power flow more accurately, we adopt a convex relaxation of the DistFlow model using second-order cone formulations:

$$\begin{aligned} V_j(t)^2 = V_i(t)^2 - 2(R_{ij} P_{ij}(t) + X_{ij} Q_{ij}(t)) \\ + (R_{ij}^2 + X_{ij}^2) I_{ij}(t) \end{aligned} \tag{33}$$

$$P_{ij}(t)^2 + Q_{ij}(t)^2 \leq V_i(t)^2 I_{ij}(t) \tag{34}$$

where $V_j(t)$ is the voltage magnitude at node j; R_{ij} and X_{ij} are the line resistance and reactance, respectively; $P_{ij}(t)$ and $Q_{ij}(t))$ are the active and reactive power flow on line ij, respectively; $I_{ij}(t)$ is the squared current magnitude.

3) Network Constraints: The network constraints include line balance constraints, line capacity constraints, and node voltage limits.

$$\sum_{k \in \mathcal{N}_i^-} P_{ki}(t) - \sum_{j \in \mathcal{N}_i^+} P_{ij}(t) = P_g^{(i)}(t) + LS_i(t) + P_{net}^{(i)}(t) \tag{35}$$

$$\sum_{k \in \mathcal{N}_i^-} Q_{ki}(t) - \sum_{j \in \mathcal{N}_i^+} Q_{ij}(t) + Q_g^{(i)}(t) = Q_i^{\text{load}}(t) \tag{36}$$

$$P_{ij}(t)^2 + Q_{ij}(t)^2 \leq (S_{ij}^{\max})^2 \tag{37}$$

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$$V_i^{\min} \le V_i(t) \le V_i^{\max}, \quad \forall i, t \tag{38}$$

where $\mathcal{N}_i^-$ and $\mathcal{N}_i^+$ are the set of upstream and downstream neighbors of node i; $S_{ij}^{\max}$ is the thermal rating of line; $V_i^{\min}$ and $V_i^{\max}$ are the minimize and maximum allowed voltage magnitude at node i.

4) Controllable Unit Constraints: Generation dispatch and load shedding are restricted by their capacities:

$$0 \le P_g(t) \le s_g \tag{39}$$

$$0 \le Q_g(t) \le Q_g^{\max} \tag{40}$$

$$0 \le LS_i(t) \le P_{ac}^{(i)}(t) \tag{41}$$

where $Q_g^{\max}$ is the reactive power capacity of generator g.

C. Benders Decomposition

To address the computational complexity of the two-stage robust optimization model, we apply Benders decomposition by separating investment and operational decisions. First, the inner minimization problem under a certain scenario $\mathbf{P}_{\text{net}}$ can be denoted as the second-stage subproblem:

$$Q(\mathbf{x}, \mathbf{P}_{\text{net}}) = \min_{\mathbf{y}} \left\{ \mathbf{c}^\top \mathbf{y} \mid \mathbf{W}\mathbf{y} \ge \mathbf{h} - \mathbf{T}\mathbf{x} - \mathbf{D}\mathbf{P}_{\text{net}} \right\} \tag{42}$$

where $\mathbf{c}$ is the operational cost coefficients; $\mathbf{W}$, $\mathbf{T}$, and $\mathbf{D}$ are the matrices encoding the operational constraints; $\mathbf{h}$ is the right-hand side constant vector. Then, the dual problem of (42) can be formulated as:

$$Q(\mathbf{x}, \mathbf{P}_{\text{net}}) = \max_{\boldsymbol{\lambda} \ge 0} \left\{ \boldsymbol{\lambda}^\top (\mathbf{h} - \mathbf{T}\mathbf{x} - \mathbf{D}\mathbf{P}_{\text{net}}) \mid \mathbf{W}^\top \boldsymbol{\lambda} \le \mathbf{c} \right\} \tag{43}$$

where $\boldsymbol{\lambda}$ is the dual variable associated with the operational constraints. Substitute the dual value into the master problem:

$$\min_{\mathbf{x} \in \mathcal{X}} \left\{ C_{\text{inv}}(\mathbf{x}) + \max_{\mathbf{P}_{\text{net}} \subset \mathcal{U}} \max_{\lambda \in \Lambda} \lambda^\top (\mathbf{h} - \mathbf{T}\mathbf{x} - \mathbf{D}\mathbf{P}_{\text{net}}) \right\} \tag{44}$$

$$\Lambda = \{ \boldsymbol{\lambda} \ge 0 \mid \mathbf{W}^\top \boldsymbol{\lambda} \le \mathbf{c} \} \tag{45}$$

Furthermore, We define an auxiliary variable η to linearize the outer max:

$$\min_{\mathbf{x}, \eta} \quad C_{\text{inv}}(\mathbf{x}) + \eta \tag{46}$$

$$\text{s.t.} \quad \eta \ge \boldsymbol{\lambda}_n^\top (\mathbf{h} - \mathbf{T}\mathbf{x} - \mathbf{D}\mathbf{P}_{\text{net}}^{(n)}) \tag{47}$$

Each inequality defines a Benders optimality cut, where $\boldsymbol{\lambda}_n^\top$ is the dual solution associated with scenario $\mathbf{P}_{\text{net}}^{(n)}$. If the subproblem is infeasible under $\mathbf{x}^{(k)}$, a feasibility subproblem is solved:

$$\min_{\mathbf{y}} \quad \| \mathbf{W}\mathbf{y} - (\mathbf{h} - \mathbf{T}\mathbf{x}^{(k)} - \mathbf{D}\mathbf{P}_{\text{net}}^{(n)}) \|_2 \tag{48}$$

If infeasibility is confirmed, a feasibility cut is generated to eliminate $\mathbf{x}^{(k)}$ from the feasible region.

V. Case Study

To evaluate the effectiveness of the proposed robust planning model, numerical experiments are conducted on a modified IEEE 33-bus distribution system. The test system includes multiple types of components affected by meteorological conditions: rooftop photovoltaic units, wind turbines, and building cooling loads. Historical meteorological datasets are collected from the China Meteorological Administration for a representative subtropical city in Eastern China, which are used to train the SARMA–LSTM hybrid model to simulate multi-timescale heatwave scenarios. The proposed model is implemented in Matlab 2024a and solved using the Gurobi Optimizer 10.0 on a workstation with an Intel Core i7-8550U CPU and 16 GB RAM. To assess the performance of proposed framework, the following benchmark strategies are compared:

(i) Stochastic Planning (SP): Multi-scenario optimization based on historical weather variations.

(ii) Proposed Robust Planning (RP): Full two-stage robust model incorporating both seasonal and long-term heatwave evolution via the SARMA–LSTM framework.

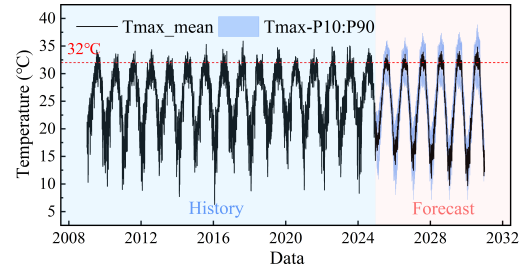

Fig. 1. The daily maximum temperature in 2009-2023.

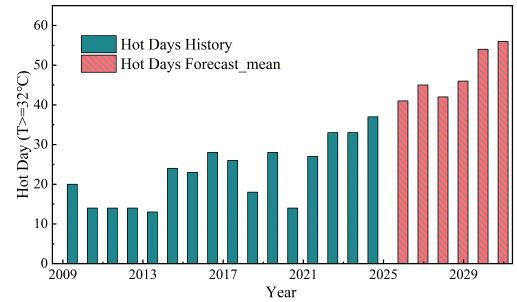

Fig. 2. The annual number of hot days in 2009-2023.

A. Multi-Timescale Simulation of Urban Heatwaves

Fig. 1 illustrates the historical (2009–2024) and projected (2025–2030) daily maximum temperature profiles, together with the 10th–90th percentile prediction bands. The historical segment demonstrates a pronounced seasonal oscillation with gradually intensifying summer peaks. In the forecast period, the daily temperature series not only maintains the inherent seasonality but also exhibits an upward shift in both mean and extreme values, thereby reflecting the progressive enhancement of summer thermal stress. The percentile envelope further indicates that the inter-annual variability is expected to expand in the future, which is aligned with the physical understanding of climate-driven extremes.

Fig. 2 quantifies the annual number of hot days (defined as $T_{\max} \ge 32°\text{C}$) based on both historical records and forecasts. A significant inter-annual variability is observed in the historical period, ranging between 10 and 40 days per year. More importantly, the forecast results present an overall

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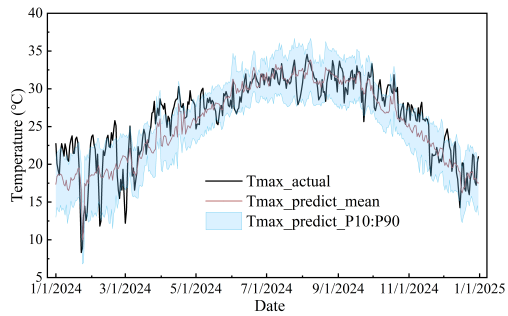

Fig. 3. The predicted and actual daily maximum temperatures for 2024.

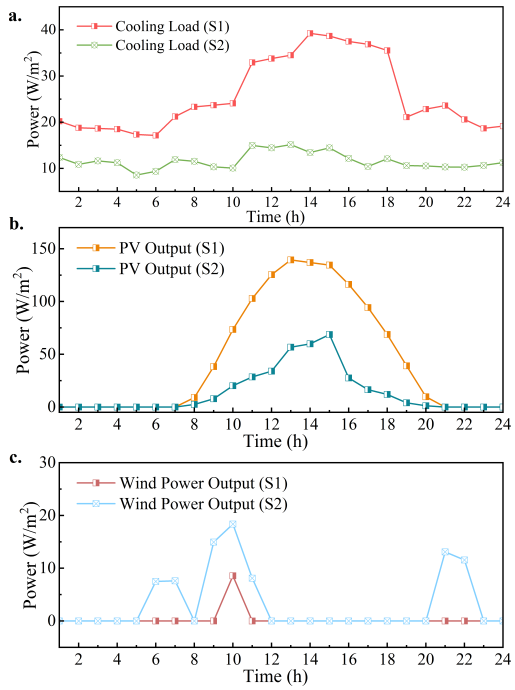

Fig. 4. The component-level response in two scenarios (S1 and S2).

increasing trend, with the mean number of hot days rising from approximately 30–40 days in 2024 to over 50 days by 2030. This monotonic escalation highlights the long-term intensification of thermal extremes and points to a progressively aggravated imbalance between load and generation in the distribution system during heatwave seasons.

Fig. 3 provides a detailed comparison between the predicted and actual daily maximum temperatures for 2024, serving as a validation of the proposed SARMA–LSTM-based forecasting framework. The predicted mean trajectory closely follows the seasonal pattern of the observed data, while the prediction envelope covers most of the actual fluctuations, including extreme spikes in summer. This agreement substantiates the effectiveness of the forecasting model in capturing both the long-term seasonal trend and the short-term variability.

B. Component-Level Response Under Heatwave Conditions

To further reveal the influence of meteorological conditions on source–load interactions, two representative scenarios in August 2024 are selected: a hot sunny day S1 (August 5) and a relatively mild cloudy day S2 (August 15). Fig. 4a–4c illustrate the hourly energy density variations of building cooling load, PV generation, and wind power under these two

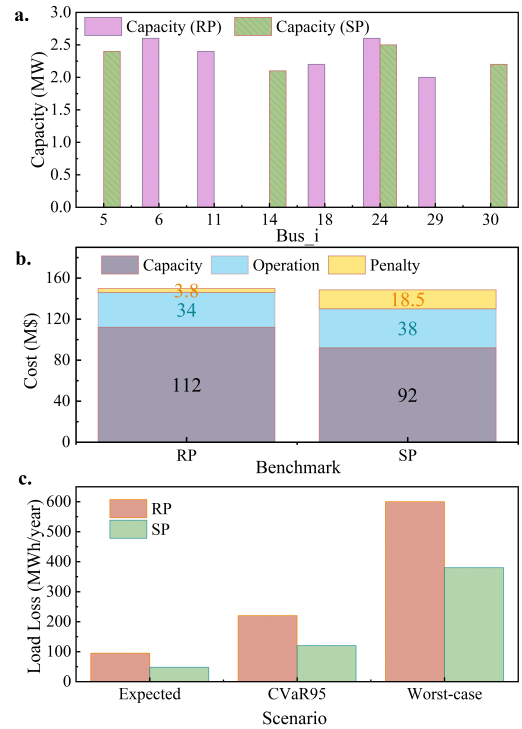

Fig. 5. The comparative analysis of two planning strategies: a.The sitting and sizing of new generation; b. the total cost of two strategies; c. the amount of load loss in expected value, CVaR95, and worse-case scenarios.

scenarios. As shown in Fig. 4a, the maximum temperature on S1 reached 34.6°C, leading to significantly higher cooling demand compared with S2. The cooling load density on S1 remained elevated from 10:00 to 18:00, with a peak about 25–30% higher than that of S2 and with an extended duration of high load. In contrast, S2 exhibited a lower peak and a much shorter period of sustained demand. This indicates that extreme heat not only elevates peak cooling requirements but also prolongs their persistence, thereby intensifying stress on the distribution network.

The comparison of PV generation is presented in Fig. 4b. The average solar irradiance on S1 was nearly three times that of S2, resulting in a much higher peak output density. However, the PV module temperature on S1 rose to 55–60 °C at noon, causing an efficiency drop of about 5–7% and leading to deviation from the ideal output curve. In S2, the module temperature remained below 40 °C, the efficiency was stable, and the output followed the irradiance profile almost linearly. These results reveal that under hot conditions, PV output is higher in magnitude but suffers from efficiency degradation and reduced stability. Moreover, as shown in Fig. 4c, during S1, the average wind speed was only 2–3 m/s, mostly below the rated operating region, resulting in relatively low output. In contrast, S2 experienced stronger winds of 4–5 m/s, enabling more stable turbine operation and approximately 30–40% higher output density compared with S1.

C. Comparative Analysis of Planning Strategies

As shown in Fig. 5a, the SP solution installs four units at buses {5, 14, 24, 30} with capacities totalling 9.2 MW. In

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contrast, the RP solution deploys five units at {6, 11, 18, 24, 29} with capacities totalling 11.8 MW. Both plans identify bus 24 as a pivotal location, while RP further shifts capacity toward the feeder's backbone (buses 6, 11) and downstream sections (buses 18, 29). By modestly upsizing and spreading capacity over topologically influential buses, RP mitigates the stress of heat-driven peak demand and renewable derating observed in the historical record.Fig. 5 b further depicts the cost distributions of the two strategies, including capacity, operation, and load shortage penalty. It can be observed that compared to SP, RP allocates more generation capacity to enhance the system's resilience against adverse scenarios. With sufficient dispatchable units, both the operating costs and the costs associated with load shortfalls decrease. Fig. 5c quantifies the adequacy improvement. Relative to SP, RP reduces the load loss from 95 to 48 MWh per year, lowers the 95th-percentile conditional value-at-risk from 220 to 120 MWh per year, and decreases the worst-case annual energy not served from 600 to 380 MWh. These outcomes indicate not only a better average performance but also a materially thinner tail of the shortage distribution, which is consequential for distribution operators during heatwave stress when peak demand and renewable derating coincide. Taken together, Fig. 5a–5c show that the robust plan reallocates and modestly upsizes capacity at a handful of structurally important buses, which translates into substantial improvements in both average adequacy and tail-risk metrics, while keeping total cost essentially on par with the stochastic plan.

VI. Conclusion

This paper presents a two-stage robust planning framework for urban distribution systems under intensifying heatwaves. By integrating SARMA–LSTM-based multi-timescale climate modeling with physics-informed source–load models, the proposed method captures both seasonal fluctuations and long-term warming trends. Embedding these uncertainties into a robust optimization model solved via Benders decomposition enables tractable planning under high-dimensional scenarios. Case studies on the IEEE 33-bus system show that the robust plan modestly reallocates and upsizes capacity at critical nodes, effectively reducing both expected and extreme shortage risks while maintaining comparable costs to stochastic planning. These findings highlight the importance of explicitly accounting for compound heatwave uncertainties in long-term distribution planning. Future work will extend the framework to include storage and demand response.

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